Ericsson PE Node JAY-PE-09 - Routing Protocol Flap Resolution Procedure

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_9916

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on the Ericsson PE Node JAY-PE-09 at the Jayanagar Metro Site is primarily attributed to the complex interplay between the BGP, OSPF, and ISIS protocols. The PE Node, being a critical component in the BGP routing domain, is responsible for maintaining a stable routing topology by exchanging routing information with its peers. However, the protocol interactions can sometimes lead to oscillations or flapping, resulting in routing instabilities. This phenomenon can be exacerbated by factors such as misconfigured routing policies, hardware issues, or software bugs. Furthermore, the use of ISIS as a backbone protocol in the Metro Site network adds an additional layer of complexity, as LSP flooding can cause network instability. Therefore, a comprehensive approach is necessary to identify and resolve the root cause of the Routing Protocol Flap failure on the Ericsson PE Node JAY-PE-09.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that all necessary physical connections are intact.

- Check the status of the redundant links and ensure that they are operational.

- Review the site's environmental conditions to ensure that they are within the recommended operating parameters.

- Confirm that the dispatch team has been notified and is en route to the site.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policies	High
Hardware issue with the PE Node	Medium
ISIS LSP flooding due to network instability	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP summary information:, Description: This command displays a summary of the BGP routing information, including the number of peers, routes, and updates. It helps to identify any issues with the BGP protocol, such as routing instabilities or flapping.

Command: show ospf database, Expected Output: OSPF database information:, Description: This command displays the OSPF routing
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information, including the link-state database and the routing table. It helps to identify any issues with the OSPF protocol, such as adjacency loss or routing instability.

Command: show isis database, Expected Output: ISIS database information:, Description: This command displays the ISIS routing information, including the link-state database and the routing table. It helps to identify any issues with the ISIS protocol, such as LSP flooding or routing instability.
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Resolution Steps

Step 1: Step: Verify BGP routing policies, Command: show bgp policy, Description: This command displays the BGP routing policies, including any configured routes and filters. It helps to identify any misconfigured policies that may be causing the Routing Protocol Flap.

Step 2: Step: Check PE Node hardware, Command: show hw status, Description: This command displays the hardware status of the PE Node, including any error messages or warnings. It helps to identify any hardware issues that may be causing the Routing Protocol Flap.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp, Description: This command clears the ISIS LSP flooding, which can help to resolve the routing instability caused by the LSP flooding.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP summary information:, Description: This command displays a summary of the BGP routing information, including the number of peers, routes, and updates. It helps to verify that the BGP protocol is stable and that the Routing Protocol Flap has been resolved.

Command: show ospf database, Expected Output: OSPF database information:, Description: This command displays the OSPF routing information, including the link-state database and the routing table. It helps to verify that the OSPF protocol is stable and that the Routing Protocol Flap has been resolved.

Command: show isis database, Expected Output: ISIS database information:, Description: This command displays the ISIS routing information, including the link-state database and the routing table. It helps to verify that the ISIS protocol is stable and that the Routing Protocol Flap has been resolved.
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Roll-back Procedures

- Step: Restore BGP routing policies, Command: apply bgp policy, Description: This command restores the original BGP routing policies, which may have been modified during the resolution process.

- Step: Re-enable PE Node hardware, Command: enable hw, Description: This command re-enables the PE Node hardware, which may have been disabled during the resolution process.

- Step: Re-create ISIS LSP flooding, Command: create isis lsp, Description: This command re-creates the ISIS LSP flooding, which may have been cleared during the resolution process.